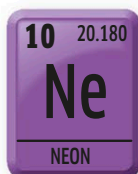
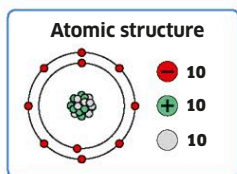


NEON

Neon

Discovered: 1898

Neon might be the most well-known of the noble gases because of its use in bright advertising signs and lighting. Like the other members of the noble gases group, it is inert (doesn't react with other elements), and quite rare. Neon is present in air in small quantities; in fact, air is the only source of this element. To extract the neon, air is cooled until it becomes liquid. Then it is heated up again and, through distillation, the different elements present in air can be harvested as they vaporize. Neon can be used as a refrigerant and, when combined with helium, it can be used in lasers.



Neon red?

When electricity is passed through neon gas, it glows a stunning red. In fact, only red neon signs are actually made of neon. Other "neon" colors come from other noble gases—argon, for example, gives blue colors.

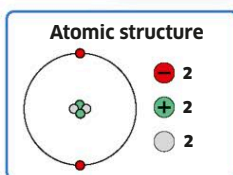
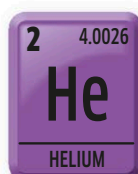


HELIUM

Helium

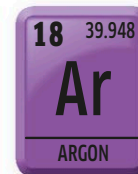
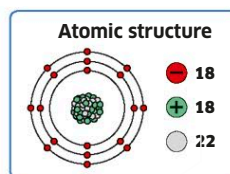
Discovered: 1895

Helium is a very light gas; only hydrogen is lighter. That is why it is put in all kinds of balloons. Airships, weather balloons, and party balloons can all be filled with the gas to make them rise and remain in the air. Helium is very unreactive. Because of this, it forms few compounds. Like neon, it can also be used as a cooling agent.



Welding flame

Argon gas is used in welding to prevent water vapor and oxygen gas reacting with the metal.



ARGON

Argon

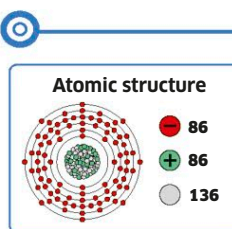
Discovered: 1894

After nitrogen and oxygen, argon is the third most abundant gas in Earth's atmosphere. It is unreactive in nature, and doesn't conduct heat very well. Its name, from the Greek word *argos*, even means "idle." It can be put to good use, however—for example in welding and to protect fragile museum artifacts from decaying in oxygen-rich air.



Dying star

The Crescent Nebula is made of gases thrown off by a dying star. Most of what remains of the star is helium, produced by millions of years of nuclear fusion.



RADON

Radon

Discovered: 1900

Radon is a colorless gas which is released from minerals in the ground that contain the element uranium. Dangerously radioactive, radon can be a serious health risk. Breathing it in can cause lung cancer. It is present everywhere, but usually at very low levels. In areas where higher levels of radon are likely, home radon testing kits are sometimes provided.



Volcanic mud

Radon is present in volcanic springs and the mud surrounding them. Scientists often monitor the levels to make sure the groundwater in the area is safe to drink.